

Excel Data Analysis – Tasks Performed

1. Removed duplicate records from the customer address data and standardized state names.
2. Cleaned customer demographic data by correcting erroneous entries, standardizing missing-data formats, and resolving inconsistencies in gender representation.
3. Cleaned transaction data by standardizing the `transaction_date` format and removing records with missing or incomplete information.
4. Standardized new customer data by formatting addresses consistently, ensuring consistent gender representation, and correcting anomalies in `job_title` and `job_industry_category`.
5. Performed customer segmentation by analyzing customer counts and average tenure across different `wealth_segment` categories.
6. Analyzed customer segments by gender by calculating customer counts and average `past_3_years_bike_related_purchases`.
7. Analyzed customer distribution across `job_industry_category` and examined the distribution of `wealth_segment` within each industry.
8. Analyzed monthly sales trends by calculating total sales per month and identifying seasonal patterns and significant sales spikes.
9. Evaluated product performance by analyzing total sales by brand and calculating total sales and average `list_price` by `product_line`.
10. Analyzed customer purchase behavior by identifying the top 10 customers based on total transaction value and calculating the average number of purchases per customer.
11. Analyzed new customer demographics by examining the distribution of new customers across `wealth_segment` and `job_industry_category`, along with their average `past_3_years_bike_related_purchases`.
12. Performed new customer location analysis by visualizing customer distribution by state and analyzing the relationship between `property_valuation` and `wealth_segment`.
13. Estimated potential revenue from new customers using `past_3_years_bike_related_purchases` and customer value.
14. Calculated Customer Lifetime Value (CLV) for individual customers using Average Purchase Value, Purchase Frequency, and Customer Lifespan based on the specified CLV formula.

15. Performed CLV segmentation by comparing average CLV across `wealth_segment` categories and analyzing the relationship between CLV and customer demographics such as gender and job industry.
16. Consolidated key findings from customer segmentation, transaction analysis, new customer insights, and CLV analysis into an executive summary.
17. Developed marketing recommendations based on the characteristics of high-value customer segments.
18. Identified potential business expansion areas using new customer location analysis.
19. Proposed product improvement opportunities based on transaction and product performance analysis.